

**Smoke Pit Final Report: Stove Cooking and Operations Optimization**  
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## 1. Executive Summary

This report presents the final simulation analysis and recommendations for the Smoke Pit project after incorporating stove cooking and finite side inventory into the restaurant model. In the updated model, baked beans, green beans, and mac and cheese are no longer treated as unlimited items. Instead, these sides must be cooked on the stove, cleaned after preparation, stored in the side cabinet, and replenished to the assembly station before they can be served.

The objective of the analysis was to identify a feasible operating policy that maximizes expected daily profit while maintaining acceptable customer-service performance. The main decision variables considered in the analysis included staffing levels, initial meat inventory, replenishment points, side assembly capacities, and side cabinet allocations.

The final recommended operating policy uses **2 customer-service workers** and **4 food-preparation workers**. Under this policy, the system produces an estimated **average daily profit of \$1,707.66**, while maintaining an **average order-to-delivery time of 6.03 minutes** and an **average total time in the system of 14.52 minutes**. This policy was selected because it produced the best overall balance of profitability and service performance among the high-performing cases considered in the final analysis.

The final recommendation was not selected only because it produced the highest original OptQuest profit. A strong operating policy must also avoid excessive order-to-delivery time, excessive balking or reneging, high worker utilization, stockout risk, and unnecessary cooked-side leftovers. Refer to Appendix A Table 1 for further information.

## 2. Model Updates, Inputs, and Optimization Design

The main difference between this final model and the earlier deliverable is that side inventory is now finite. In earlier modeling stages, sides were effectively assumed to be always available. In the updated model, baked beans, green beans, and mac and cheese must first be cooked on the stove and then stored in the side cabinet before they can be moved to the assembly station. This change creates a much more realistic restaurant system because it introduces the possibility of side shortages, replenishment delays, and food waste. If too little is cooked, stockouts and lost sales increase. If too much is cooked, side cost and leftovers increase.

The model includes two levels of replenishment:

1. **Side cabinet replenishment from stove cooking**
2. **Side assembly replenishment from the side cabinet**

This structure allows the model to capture the operational delay between deciding to cook more sides and actually making those sides available to the customer.

## Input Distributions

The cooking and cleaning times for side items were fitted using Simio's Input Analyzer based on the Smoke Pit time-study data. Weibull distributions were selected because they provided a reasonable fit and are appropriate for nonnegative task times with some right-skewness.

Also, because the model uses **BeanPrepDist** and **BeanCleanDist**, the same bean preparation and cleaning distributions were applied to **both baked beans and green beans**. **Decision Variables Considered**

The main decision variables optimized in the model were:

- Number of customer-service workers
- Number of food-preparation workers
- Initial pork, brisket, and rib trays
- Meat replenishment points
- Side assembly capacities
- Side replenishment points
- Fry replenish point
- Side cabinet capacities
- Stove cooking batch quantities

These controls were selected because they directly affect staffing cost, service speed, food availability, and the tradeoff between lost sales and food waste. Refer to Appendix C Table 3 for further information.

## Feasibility Constraints

Feasibility constraints were added so that the optimization procedure would not select unrealistic operating policies. These included:

- Total side cabinet capacity limit
- Side assembly feasibility
- Meat replenish feasibility
- Replenishment and storage compatibility

These constraints ensured that the final policy remained operationally realistic.

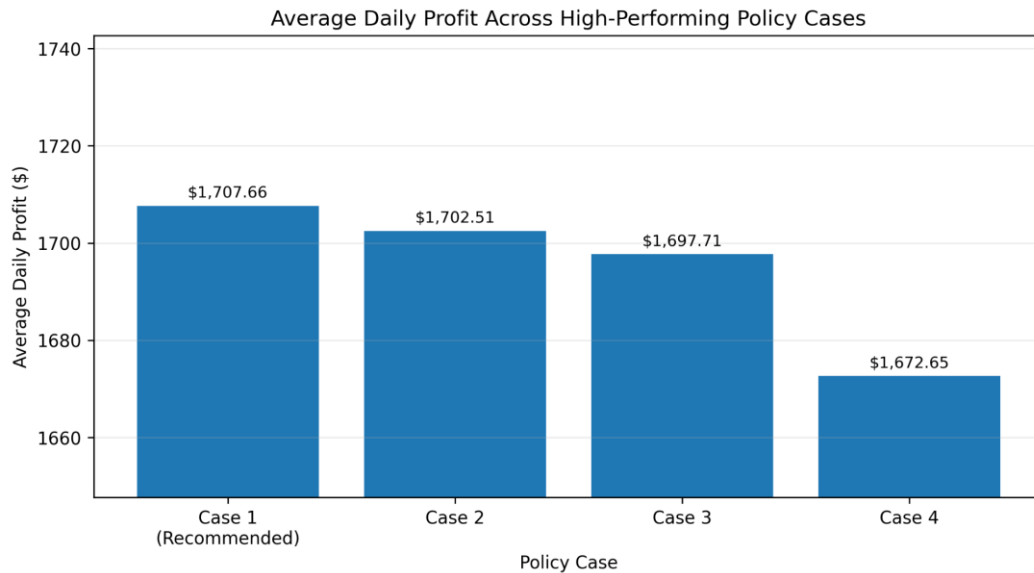
## 3. Optimization Results and Recommended Operating Policy

The final analysis compared the recommended policy against several other high-performing cases to confirm that the selected policy provides the best balance of profit and service performance. The recommended policy achieved the **highest average daily profit** among the cases compared and also maintained reasonable customer-service performance.

The selected policy uses **2 customer-service workers** and **4 food-preparation workers**. This result suggests that food-preparation capacity is more critical to overall system performance than adding additional order-taking capacity. This is reasonable because food-preparation workers are

responsible for multiple tasks, including meal assembly, meat replenishment, side assembly replenishment, and stove-cooking support.

Compared with other strong cases, the recommended policy offered the strongest overall financial performance while keeping average order-to-delivery time close to 6 minutes. Some comparison cases performed slightly better on service speed, but they did so with lower overall profitability. Other cases with fewer food-preparation workers showed lower profit and noticeably worse service performance, confirming that food-preparation staffing is a key driver of system success.



This shows the recommended policy has the highest profit.

#### 4. Additional Staffing and Expansion Analysis

To address the additional staffing-allocation and expansion questions, a short what-if analysis was performed using the recommended operating policy as the reference point. The purpose was to test whether changing food-preparation staffing or order-taking capacity would improve performance.

The results support the recommendation to use a **flexible food-preparation staff strategy**. Reducing food-preparation workers from 4 to 3 lowered profit and worsened service performance, which confirms that food-preparation capacity is a key bottleneck. Adding an extra food-preparation worker improved service time and reduced balking, but it lowered profit because of the added labor cost. Therefore, a full-time dedicated cook or replenishment worker is not recommended under the current demand level. Instead, food-preparation workers should roam flexibly between assembly, replenishment, and stove-cooking tasks, with one worker temporarily acting as a replenishment or cooking lead during peak periods.

A new order window was tested using added customer-service capacity as a proxy. This option did not improve profit compared with the recommended policy, suggesting that order-taking capacity is not the main bottleneck. The main constraints are still downstream food preparation,

side replenishment, and inventory availability. Patio expansion was not directly tested because the current model does not include a separate patio-capacity or waiting-area-capacity control. If Smoke Pit wants to evaluate patio expansion, a future version of the model should explicitly include seating or waiting-area capacity and connect it to customer balking behavior. Detailed what-if results are provided in Appendix B Table 2.

## 5. Verification, Input Uncertainty, Assumptions, and Risk

The final simulation model was checked against the approximate model developed in Project 1. The approximate model is simpler and does not include detailed stove cooking, finite side cabinet storage, side assembly replenishment, cleaning time, or shared worker responsibilities. Therefore, exact numerical agreement is not expected. However, the final model is considered reasonable because it follows the expected operational trends: lower staffing increases congestion, lower inventory increases shortages, and more preparation capacity improves service performance.

Input uncertainty remains an important consideration because the cooking and cleaning distributions were estimated from time-study data. If the available sample is limited, the fitted distributions may not fully capture all day-to-day variation. Additional time-study data would help improve the reliability of future recommendations.

**Main Assumptions:** The final model assumes that the time-study data represents normal operations, workers follow the modeled task logic, customer demand patterns remain stable, and side cabinet capacity reflects actual storage limits. The model may not fully capture informal multitasking, employee learning, worker fatigue, weather effects, special events, or unexpected demand spikes.

**Main Risks:** The main operational risk is that the recommended policy is optimized for expected daily performance. On unusually busy days, food-preparation workers and side replenishment may still become bottlenecks. The number of switched orders is also relatively high, which suggests that inventory availability continues to affect customer choices.

Overall, the recommended policy provides the best balance between profit and service performance among the final evaluated options. It improves expected daily profit while keeping average order-to-delivery time near 6 minutes and maintaining a feasible operating structure for the updated stove-cooking model.

**Final Recommendation:** Smoke Pit should implement **Option 1**, which uses **2 customer-service workers** and **4 food-preparation workers**. This policy produces the highest expected average daily profit of **\$1,707.66** while maintaining an average order-to-delivery time of approximately **6.03 minutes**. The recommendation should be monitored after implementation,

especially for side replenishment performance, food-preparation workload, and customer order switching.

### Appendix A. Final Recommendation Summary

| Decision Area                  | Recommended Value              | Main Reason                                                        |
|--------------------------------|--------------------------------|--------------------------------------------------------------------|
| Customer-service workers       | <b>2</b>                       | Provides enough order-taking capacity while controlling labor cost |
| Food-preparation workers       | <b>4</b>                       | Supports assembly, replenishment, and stove-cooking tasks          |
| Final selected scenario        | <b>Scenario 187</b>            | Highest confirmed mean profit among retained KN scenarios          |
| Expected daily profit          | <b>\$1,707.66</b>              | Primary optimization response                                      |
| Profit 95% CI                  | <b>\$1,659.65 – \$1,755.67</b> | Shows uncertainty after additional replications                    |
| Average order-to-delivery time | <b>6.03 minutes</b>            | Indicates acceptable service performance                           |
| Balked customers               | <b>62.22</b>                   | Measures customers who left before ordering                        |
| Reneged customers              | <b>7.34</b>                    | Measures customers who abandoned after waiting                     |
| Switched orders                | <b>236.30</b>                  | Measures substitution caused by inventory limits                   |

### Appendix B. Staffing and Expansion What-If Results and Input Distributions

| Case   | Description                           | CS Workers | FP Workers | Avg. Daily Profit | Avg. OTD | Avg. TIS | Balked | Reneged | Switched Orders |
|--------|---------------------------------------|------------|------------|-------------------|----------|----------|--------|---------|-----------------|
| Case 1 | Current recommended flexible policy   | 2          | 4          | \$1,663.24        | 5.92     | 14.23    | 55.65  | 10.15   | 229.25          |
| Case 2 | Lower FP capacity                     | 2          | 3          | \$1,377.74        | 8.18     | 17.51    | 108.10 | 20.10   | 182.20          |
| Case 3 | Extra FP / replenishment-cook support | 2          | 5          | \$1,448.89        | 5.11     | 13.00    | 45.65  | 9.90    | 235.20          |
| Case 4 | New order-window proxy                | 3          | 4          | \$1,366.27        | 6.11     | 14.46    | 65.20  | 7.30    | 227.35          |

### Appendix C. Selected Input Distribution

| <b>Input Process</b> | <b>Selected Distribution</b>      | <b>Applies To</b>           | <b>Justification</b>                   |
|----------------------|-----------------------------------|-----------------------------|----------------------------------------|
| Mac preparation      | <b>Weibull(16.3, 4.6) minutes</b> | Mac and cheese              | Fits positive cooking time variability |
| Mac cleaning         | <b>Weibull(7.5, 2.6) minutes</b>  | Mac and cheese              | Fits cleanup-time variation            |
| Bean preparation     | <b>Weibull(15.1, 3.7) minutes</b> | Baked beans and green beans | Shared cooking process and variability |
| Bean cleaning        | <b>Weibull(12.9, 2.8) minutes</b> | Baked beans and green beans | Shared cleanup process and variability |

### Appendix D. Supporting Graphs

